# Cheng Xu

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**Research Interests:** VLA Models, Robotic Manipulation, World Models, Multimodal Perception, and Tactile Sensing

## Education

**The University of Tokyo** — Tokyo, Japan
*M.Eng. in Electrical Engineering and Information Systems* — *Apr. 2026 – Present*
*Current research:* Vision-based landmark recognition and multi-sensor fusion for train localization in autonomous train operation.

**University of Sussex** — Brighton, United Kingdom
*B.Eng. (Hons) in Robotics and Electrical Engineering, First-Class Honours* — *Sep. 2021 – Jun. 2025*
Dual-degree program average: **88/100** | Rank: **Top 5%**

**Zhejiang Gongshang University** — Hangzhou, China
*B.Eng. in Electronic Information Engineering* — *Sep. 2021 – Jun. 2025*
Dual-degree program average: **88/100** | Rank: **Top 5%**

## Selected Experience

**AGIBOT Japan** — Tokyo, Japan
*Engineering Assistant Intern* — *Jun. 25 – Jul. 3, 2026*

- Assisted engineers with the disassembly and maintenance inspection of G2 humanoid robots during on-site technical support.
- Reviewed the operating requirements of a planned robotic towel-loading task and conducted an on-site assessment of the working environment.

## Selected Projects

**Tactile Sensors for Robotic Manipulators** — *Oct. 2024 – Mar. 2025*

- Fabricated and experimentally evaluated an eight-layer, 16×16 flexible piezoresistive tactile sensor array using Velostat as the sensing material and an Arduino Nano for row–column signal acquisition.
- Implemented a row–column readout circuit and PCB layout in EasyEDA, and integrated an Arduino–Python pipeline for serial data acquisition and real-time pressure-map visualization.
- Conducted load-response and contact-distribution experiments, compared multiple noise-reduction methods, and selected a four-stage pipeline.

**VLA Fundamentals** — *May 2026 – Present*

- Conducted a line-by-line analysis of Physical Intelligence's official $\pi_0$ PyTorch implementation, tracing tensor shapes and data flow through multimodal preprocessing, prefix and suffix embeddings, attention-mask construction, flow-matching training, and iterative action denoising.
- Ongoing work: preparing to reproduce and fine-tune $\pi_{0.5}$ based on research papers for robotic manipulation tasks.

## Technical Skills

**Programming & ML:** Python, PyTorch, C, MATLAB
**Hardware & Tools:** Arduino, EasyEDA, Multisim, schematic capture, PCB layout

## Honors & Awards

- **Outstanding Student Award (Top 1%)**, Zhejiang Gongshang University, 2024 – 2025
- **First-Class Scholarship**, University of Sussex, 2024 – 2025
- **Academic Scholarship**, University of Sussex, 2022 – 2023

## Languages

**Japanese:** JLPT N1
**English:** TOEFL iBT 87